

Landon Butler

landonb@berkeley.edu

landonbutler.github.io

Research Interests

Thesis Scalable Attribution and Selection for Machine Learning via Spectral Methods
Topics Data-centric ML, Interpretability, Trustworthy ML, Signal Processing.

Education

Exp. May 2027 **University of California, Berkeley**
Ph.D. candidate in Electrical Engineering and Computer Science
Advisor: Kannan Ramchandran
Committee: Bin Yu, Alex Dimakis

May 2022 **University of Pennsylvania**
M.S.E. in Data Science, B.S.E. in Systems Engineering
Thesis: Convolutional Learning on Multigraphs
Advisor: Alejandro Ribeiro

Publications

IEEE BITS 26 **Spectral Sparsity: A Unifying Framework for Scalable Model Interpretability Using Codes**
Justin Singh Kang, Darin Tsui, Yigit Efe Erginbas, [Landon Butler](#), Amirali Aghazadeh, Kannan Ramchandran

ICML 26 **An Odd Estimator for Shapley Values**
Fabian Fumagalli, [Landon Butler](#), Justin Singh Kang, Kannan Ramchandran, R. Teal Witter

NeurIPS 25 **ProxySPEX: Inference-Efficient Interpretability via Sparse Feature Interactions in LLMs**
[Landon Butler](#)^{*}, Abhineet Agarwal^{*}, Justin Singh Kang^{*}, Yigit Efe Erginbas, Bin Yu, Kannan Ramchandran

ICML 25 **SPEX: Scaling Feature Interaction Explanations for LLMs**
Justin Singh Kang^{*}, [Landon Butler](#)^{*}, Abhineet Agarwal^{*}, Yigit Efe Erginbas, Ramtin Pedarsani, Kannan Ramchandran, Bin Yu

NeurIPS 24 **Learning to Understand: Identifying Interactions via the Mobius Transform**
Justin Singh Kang, Yigit Efe Erginbas, [Landon Butler](#), Ramtin Pedarsani, Kannan Ramchandran

ICASSP 24 **Non-Commutative Convolutional Signal Models in Neural Networks: Stability to Small Deformations**
Alejandro Parada-Mayorga, [Landon Butler](#), and Alejandro Ribeiro

IEEE TSP 23 **Convolutional Learning on Multigraphs**
[Landon Butler](#), Alejandro Parada-Mayorga, and Alejandro Ribeiro

IEEE TSP 23 **Convolutional Filters and Neural Networks with Non-Commutative Algebras**
Alejandro Parada-Mayorga, [Landon Butler](#), and Alejandro Ribeiro

ICASSP 23 **Learning with Multigraph Convolutional Filters**
[Landon Butler](#), Alejandro Parada-Mayorga, and Alejandro Ribeiro

CEUS 23 **Equitable Optimization of U.S. Airline Route Networks**
Andy Eskenazi, Arnav Joshi, [Landon Butler](#), and Megan Ryerson

ICRAT 22 **Democratizing Aviation Emissions Estimation: Development of an Open-Source, Data-Driven Methodology**
Andy Eskenazi, [Landon Butler](#), Arnav Joshi, and Megan Ryerson

IROS 21 **Learning Connectivity for Data Distribution in Robot Teams**
Ekaterina Tolstaya^{*}, [Landon Butler](#)^{*}, Daniel Mox, James Paulos, Vijay Kumar, and Alejandro Ribeiro

Experience

Fall 2026	Google Deepmind — Student Researcher Topic: Interpretability for AI Weather Prediction.	<i>London, UK</i>
Summer 2026	Snorkel AI — Research Intern Topic: Data valuation for frontier reasoning models.	<i>San Francisco, CA</i>
Summer 2025	Apple — Machine Learning Intern Topic: Machine learning for communication systems.	<i>San Diego, CA</i>
Summer 2024	Uber — Software Engineering Intern Topic: Embedding models for search and retrieval.	<i>Sunnyvale, CA</i>

Teaching

Fall 24	Signals and Systems, <i>University of California, Berkeley</i>
Fall 21	Graph Neural Networks, <i>University of Pennsylvania</i>
Fall 21	Foundations of Data Science, <i>University of Pennsylvania</i>
Spring 21	Statistics for Data Science, <i>University of Pennsylvania</i>

Fellowships

2022	Graduate Research Fellowship, <i>National Science Foundation</i>
2021	Littlejohn Fellowship, <i>University of Pennsylvania</i>